



NFR Traceability Matrix

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David van Rooijen

Dandré Nel

Daniel Cohen

Stephan Kritzinger

Micheal Koch



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ID	Quantified Requirement	Tactic in SAS	Test / Tool	Target / Actual
NFR-1	Sensitive user data encrypted (AES-256/TLS1.3), RBAC enforced on API endpoints	AES-256/TLS1.3 encryption and the use of JWT with RBAC to enforce API endpoint access	Mozilla HTTP Observatory Badge, Bandit, pip-audit, npm audit	Observatory: A+/A+ Bandit: 0 High (Severity) / 0 High Severity pip-audit: 0/0 npm-audit: 0/0
NFR-2	>= 99.9% uptime	Docker Compose restart policies ensure automatic recovery of crashed containers, PWA functionality ensures the application still works even if the server becomes temporarily unavailable	UptimeRobot Badge	>= 99.9% / 96.4% (Read NFR-4 for reasons)
NFR-3	Respond within 2s for 95% of requests, support >= 60 concurrent users	FastAPI's async I/O prevents blocking during concurrent requests, PostGIS GiST indexing and Redis caching improve performance during heavy queries, MapLibre GL handles client-side rendering	k6 performance tests, approximately 60 concurrent users, running on every merge to ensure performance	p95 <= 2s / p95 <= 1.4s

NFR-4	<= 2 hours downtime per month	CI/CD prevents broken code from reaching production, redeploy previous-commit strategy ensures rapid recovery within 2 hours if deployment fails	Automated container restarts, UptimeRobot status badge	<= 2h per month / ~24h due to github actions outage leading to a failing deployment. Actions have been taken to remedy this for the future
NFR-5	New features/bug fixes deployable within 2 hours	Same CI/CD + redeploy tactic as NFR-5	GitHub Actions CI must pass (regression gate) and measured pipeline duration (commit -> CI pass -> deploy complete) via Actions run history	All tests pass and <=2h wall-clock / Max action time is ~10m
NFR-6	>= 75% automated test coverage	GitHub Actions run an automated test suite, Codecov ensures test coverage hits at least 75%	Unit and Integration tests, End-to-End tests, Backend CI Badge, Frontend CI Badge, Codecov Badge	>= 75% / 91%